

# Cloud Architecture for Sofa-so-good: A Personal Data Engineering Journey

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This article details the process of designing and implementing a cloud-based data architecture for Sofa-so-good, a medium-sized furniture retail company. As part of a personal data science project, I designed a scalable and reliable cloud infrastructure to streamline the company's data access, analytics, and reporting capabilities. This report outlines the design decisions, technical components, and business applications of the final architecture.

## 1. Introduction

The growing importance of data in decision-making has made modern data infrastructures an essential part of business success—even for mid-sized enterprises. As a Data Science student, I undertook a project to create a cloud architecture for Sofa-so-good, a furniture retail company with five physical locations and a thriving e-commerce store. While the company processes around 100 online orders per day—making its data volume manageable—it lacked a unified, efficient data pipeline for analysis and reporting.

The goal of this project was to build a robust cloud-based solution to collect, process, and analyze data, enabling real-time insights and future scalability.

## 2. Project Objective and Purpose

### Objective

To design a data infrastructure that centralizes and streamlines business data for Sofa-so-good, enabling analytics, forecasting, and reporting from a single, accessible source.

**Purpose**

- Centralize and integrate disparate data sources.
- Automate data processing and transformation.
- Enable easy data retrieval for advanced analytics and reporting.
- Provide a single source of truth for all business data.

**Mission:**

To design and implement a high-performance, reliable data engineering pipeline that ensures accurate, readily accessible, and consistently available data, empowering Sofa-so-good with a foundational infrastructure for future analytics and operational improvements.

**Goals:**

- Establish a centralized data repository to streamline access to information and facilitate data-driven decision-making across all departments.
- Develop a robust data engineering pipeline that accommodates evolving business needs, prioritizing process optimization, cost control, and data availability.

### 3. System Design and Planning

Before to start building the cloud architecture it was necessary to understand how the business works, which data sources they have, how can it be used and by who.



#### 3.1. Data Inputs

##### a. Sales Transactional Data:

- Nature: Detailed records of in-store and point-of-sale transactions, including product details, prices, quantities, dates, customer information, and payment methods.
- Format: CSV, relational database tables (e.g., PostgreSQL, MySQL).
- Ingestion type: Batch ingestion.
- Usage: Sales analysis, forecasting, customer behavior analysis, inventory management, and financial reporting.

##### b. E-Commerce Data:

- Nature: Online sales data, including product views, cart additions, checkout processes, order details, customer accounts, and online payment information.
- Format: JSON, relational database tables.
- Ingestion type: Stream ingestion for real-time order processing, and batch ingestion for daily/weekly reports.
- Usage: Online sales analysis, customer journey tracking, website conversion optimization, and personalized online marketing.

c. Website Logs:

- Nature: Detailed records of user interactions on the Sofa-so-good website, including page views, clicks, search queries, session durations, and device information.
- Format: JSON, log files.
- Ingestion type: Stream ingestion for real-time web traffic analysis.
- Usage: Website analytics, user behavior analysis, A/B testing, and marketing campaign performance evaluation.

d. Warehouse Operational Data:

- Nature: Data related to warehouse inventory, stock movements, shipping, receiving, and order fulfillment processes.
- Format: CSV, Excel, database tables, API feeds from Warehouse Management Systems (WMS).
- Ingestion type: Batch ingestion for daily/weekly inventory updates.
- Usage: Inventory management, stock optimization, order fulfillment tracking, and supply chain analysis.

### **3.2. Data Outputs**

a. Sales Performance Dashboards:

Near real-time sales tracking, product performance, and regional sales analysis.

User needs: Sales and management teams for monitoring sales trends and performance.

b. Inventory Management Reports:

Stock levels, reorder points, and inventory turnover rates.

User needs: Operations and inventory management teams for optimizing stock levels.

c. Customer Segmentation Reports:

Customer demographics, purchase behavior, and personalized recommendations.

User needs: Marketing and sales teams for targeted campaigns and customer relationship management.

d. Sales Forecasting Reports:

Predictive analysis of future sales based on historical data and market trends.

User needs: Management and sales teams for resource allocation and planning.

e. Web Analytics Dashboards:

Website traffic, user engagement, and conversion rates.

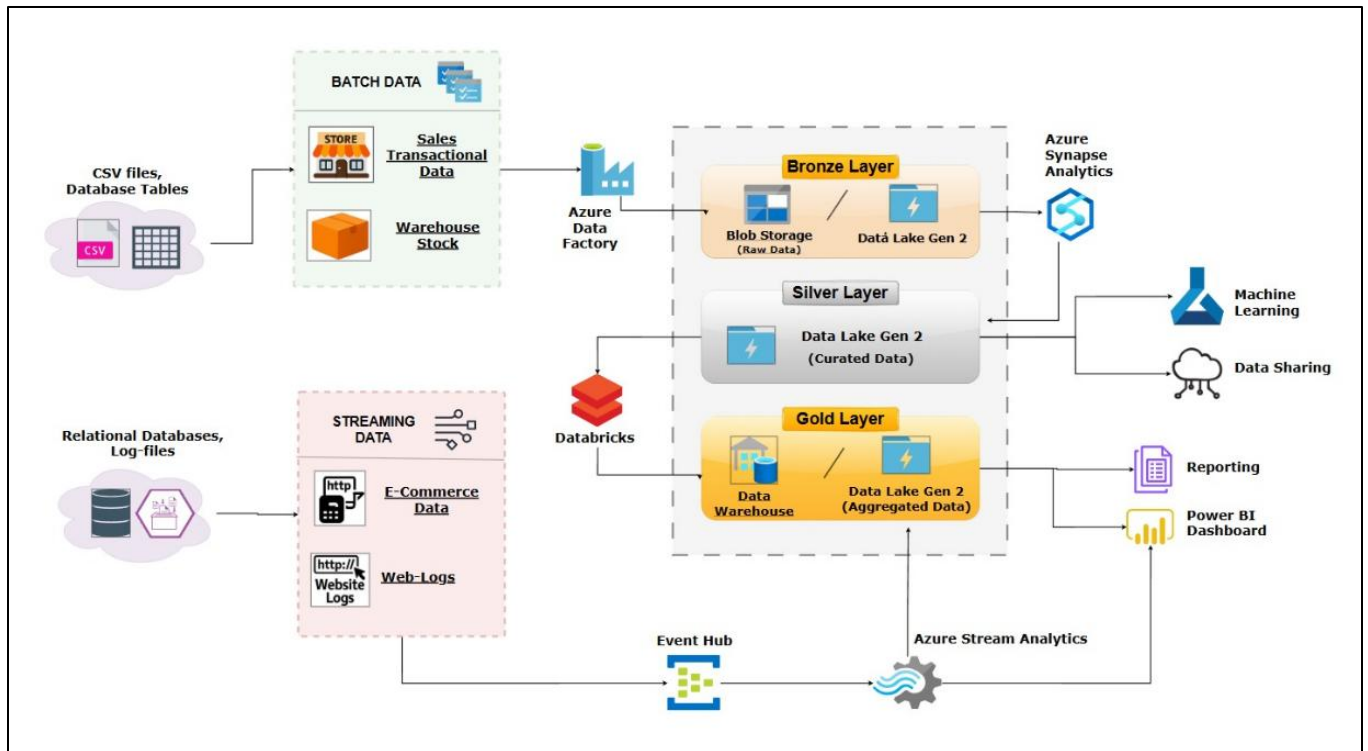
User needs: Marketing and web development teams for optimizing website performance.

### **3.3. List of Data Users**

- Marketing Team: Uses customer segmentation, web analytics, and sales data for targeted campaigns and personalized marketing.
- Sales Team: Uses sales performance dashboards, customer purchase history, and forecasting reports to improve sales strategies and customer interactions.
- Operations Team: Uses inventory management reports and supply chain analysis to optimize stock levels and warehouse operations.
- Management Team: Uses sales performance dashboards, forecasting reports, and overall business analytics for strategic decision-making.
- Customer Service Team: Uses customer data and purchase history to provide personalized and efficient customer support.
- Web Development Team: Uses web analytics and API data to improve website functionality and customer experience.

## 4. Cloud Architecture Design

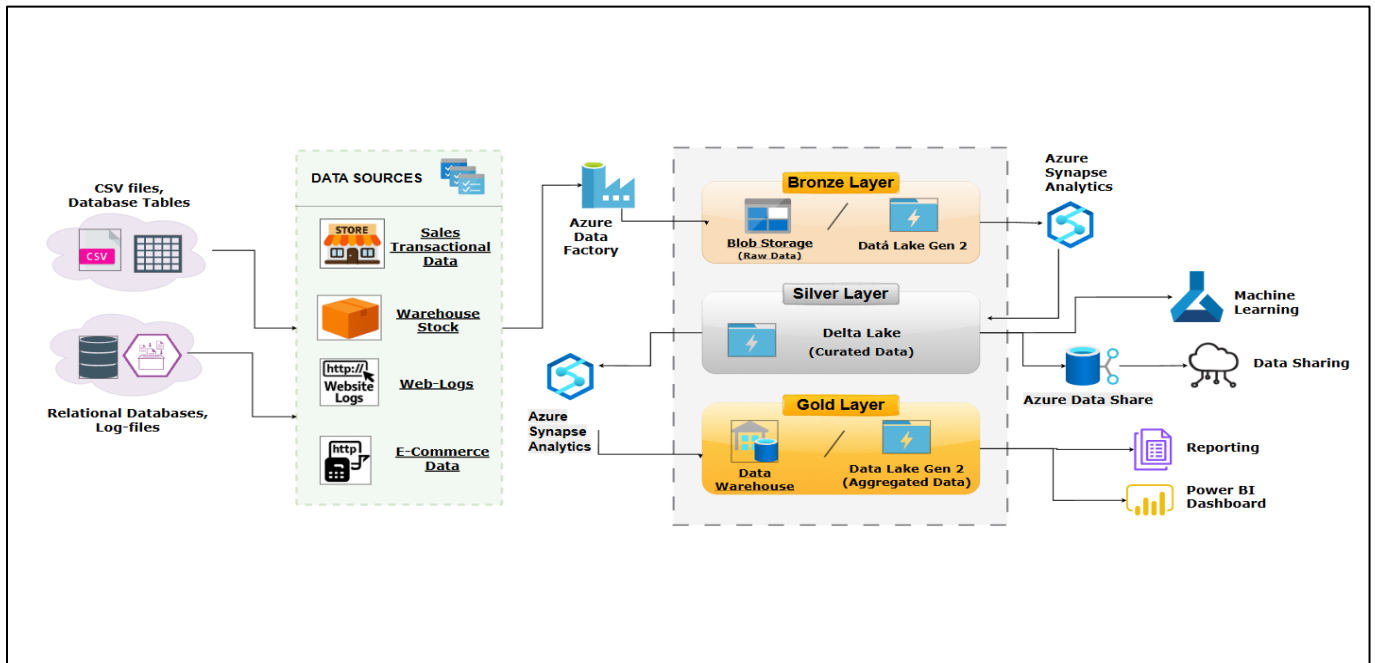
### Phase 1: Initial Design



In the first phase I considered using two different flows, one for streaming data and one for batch data. Azure Event Hub was going to handle the streaming from E-commerce and Website traffic logs and Azure Data Factory was going to handle the batch ingestion from the stores transaction data and warehouse operations.

This design would allow us to handle all data appropriately and be able to have real-time insights from the stream ingested data.

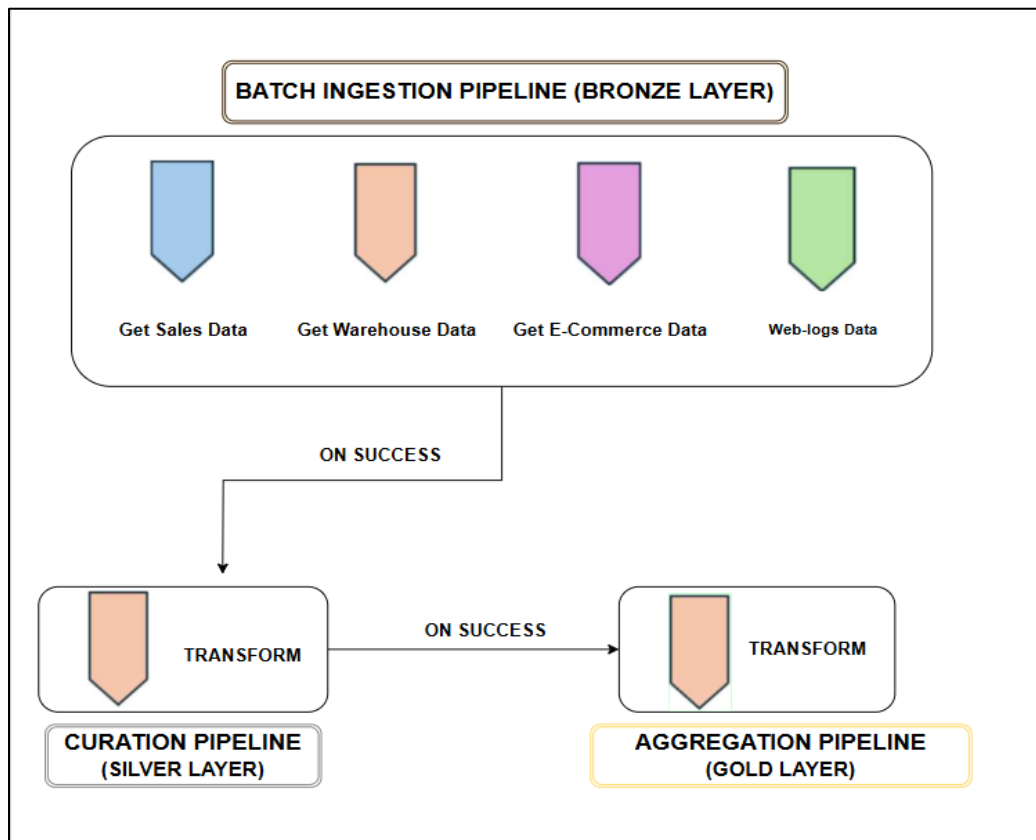
## Phase 2: Improved Design



After more discussions with Sofa-so-good leadership, I determined that having access to exact real-time data would not provide additional benefits to Sofa-so-good, compared to having the data with a 1-1.5 hour delay. For this reason, I updated the architecture design to eliminate the ingestion of streaming data using Azure Event Hub. This solution will be more cost-effective for the company's operations.

Now, all the data will be ingested by Azure Data Factory as a batch every hour. Then the data will follow the data flow from Bronze layer (raw data) to Silver layer (cleaned) to Gold layer (aggregated data)

## 5. Pipeline Design



The attached image illustrates the batch data ingestion pipeline designed with a multi-layer architecture. In the Bronze Layer, raw data is ingested from four distinct sources: Sales Data, Warehouse Data, E-Commerce Data, and Web-logs Data. Upon successful ingestion, the data proceeds to the Silver Layer, labeled as the Curation Pipeline, where an initial transformation process takes place. Following a successful transformation in the Silver Layer, the data moves to the Gold Layer, identified as the Aggregation Pipeline, where a further transformation, involving aggregation and summarization, is performed to prepare the data for analysis and reporting. This layered approach allows for data quality management and specific transformations at each stage of the pipeline.

## 6. Conclusion and Future Plans

In conclusion, the developed cloud architecture for Sofa-so-good represents a strong balance between efficiency and adaptability. The layered pipeline, with its Bronze, Silver, and Gold layers, optimizes data flow and transformation, ensuring that data is processed in a structured and efficient manner. This design facilitates streamlined data ingestion from diverse sources and supports the generation of key business reports and dashboards, ultimately enhancing operational decision-making. Furthermore, the architecture's modularity and cloud-based nature provide the adaptability needed to scale with Sofa-so-good's growing data volume and evolving business requirements. The inherent flexibility of the cloud environment allows for seamless integration of new data sources or analytical tools, ensuring the architecture remains an asset for the company's future growth.